

JEBIN KIM

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EDUCATION

M.S. Candidate in Cybersecurity, Korea University - Seoul, Republic of Korea Mar 2025 - Present
Relevant Coursework: A, B, C, and D.

B.S. in Mathematics, University of Seoul - Seoul, Republic of Korea Mar 2019 - Feb 2025
Studied number theory, linear algebra, abstract algebra, discrete mathematics GPA/Percentage : 4.12/96.2

B.E. in Artificial Intelligence, University of Seoul - Seoul, Republic of Korea Mar 2019 - Feb 2025
Studied pattern recognition, deep-learning, machine learning, data mining GPA/Percentage : 4.12/96.2

RESEARCH INTERESTS

- **Cryptanalysis of Lattice-Based Cryptography**
- **Research on AI-assisted Cryptanalysis Techniques**

RESEARCH PROJECTS

- The better Enumeration Algorithm Using Dual HKZ-reduced Basis
- Optimizing General Number Field Sieve Using Deep-Learning
- Refined Hardness Estimation of LWE

SKILLS

Frameworks/Tools	PyTorch, gymnasium, scikit-learn, pandas
Programing Languages	Python, C
Languages	Korean, English

PUBLICATIONS

Domestic Journals and Conferences (Korea)

- Jihoon Jang, Hyunju Park, Jebin Kim, Suhri Kim, Seokhie Hong, " Optimization of HQC Reed-Solomon Decoding Using AVX2 and GFNI," in Proc. of the KIISC Summer Conference (CISC-S), May. 2026.
- Jebin Kim, Insup Lee, Chanho Jeon, Suhri Kim, Seokhie Hong, and Sangjin Lee, "Reinforcement Learning for Parameter Optimization in CADO-NFS Polynomial Selection," in Proc. of the KIISC Winter Conference (CISC-W), Nov. 2025.
- Jebin Kim, Taehun Kang, Chanho Jeon, Suhri Kim, and Seokhie Hong, "A Survey on Cryptanalysis Techniques for LWE-based Cryptography," in Proc. of the KIISC Summer Conference(CISC-S), June. 2025.
- Jebin Kim, Jihoon Jang, Myeonghoon Lee, Suhri Kim, Jaechul Sung, and Seokhie Hong, "Performance Analysis of Round 2 KpqC Algorithms Considering Transmission Cost in Communication Environments," in Proc. of the KIISC Winter Conference(CISC-W), Nov. 2024.

AWARDS AND HONORS

University of Seoul

- The special prize, 2024 Cryptographic Analysis Contest, KIISC, Republic of Korea.
- Selection of Outstanding Capstone Design Projects(Finderz), University of Seoul, Republic of Korea.

TEACHING EXPERIENCE

Teaching Assistant

Mar 2023 - Feb 2025

University of Seoul

Seoul, KOR

- Teaching Assistant of Linear Algebra 2 (Sep 2024 - Feb 2025).
- Teaching Assistant of Cryptographic Mathematics 2 (Sep 2024 - Feb 2025).
- Teaching Assistant of Cryptographic Mathematics 1 (Mar 2024 - Aug 2024).
- Teaching Assistant of Analysis 2 (Sep 2023 - Feb 2024).
- Teaching Assistant of Calculus 1 (Mar 2023 - Aug 2023).
- Teaching Assistant of Linear Algebra 1 (Mar 2023 - Aug 2023).

OTHER EXPERIENCE

University of Seoul

- 2024 Advanced Cryptographic Module Professional Training Program Completion, KISA, Republic of Korea.
- Completed the 2023 Cybersecurity Academy Program for Undergraduate and Graduate Students, NIS, NSR, ICLS, Republic of Korea.